

4 Further dynamics	4.1	Newton's laws of motion, for a particle moving in one dimension, when the applied force is variable.	The solution of the resulting equations will be consistent with the level of calculus required in A level Mathematics and Further Mathematics. Problems may involve the law of gravitation, i.e. the inverse square law.
	4.2	Simple harmonic motion. Oscillations of a particle attached to the end of elastic string(s) or spring(s). Kinetic, potential and elastic energy in the context of SHM.	Proof that a particle moves with simple harmonic motion in a given situation may be required (i.e. showing that $\ddot{x} = -\omega^2 x$). Students will be expected to be familiar with standard formulae, which may be quoted without proof. e.g. $x = a \sin \omega t, \quad x = a \cos \omega t,$ $v^2 = \omega^2 (a^2 - x^2), \quad T = \frac{2\pi}{\omega}$